# SMART INDIA HACKATHON 2026

## JalRakshak

*Your Water. Your Village. Your Voice.*

- **Problem Statement ID -** SIH26255
- **Problem Statement Title -** Development of a solution using AI and ML to monitor the functionality of FHTC in real time at both the Gram Panchayat and household levels. This solution should provide alerts on water quality issues, allow village members to give feedback through mobile devices, and automatically sync data with IMIS/Sujal Gaon IDs.
- **Organization -** Ministry of Jal Shakti (MoJS)
- **Department -** National Jal Jeevan Mission (NJJM), Department of Drinking Water & Sanitation (DDWS)
- **Theme -** Clean & Green Technology
- **PS Category -** Hardware
- **Team ID -** 158724
- **Team Name -** Team Vistaar

Team Vistaar

# THE PROBLEM & OUR SOLUTION

SMART INDIA HACKATHON 2026

## THE PROBLEM

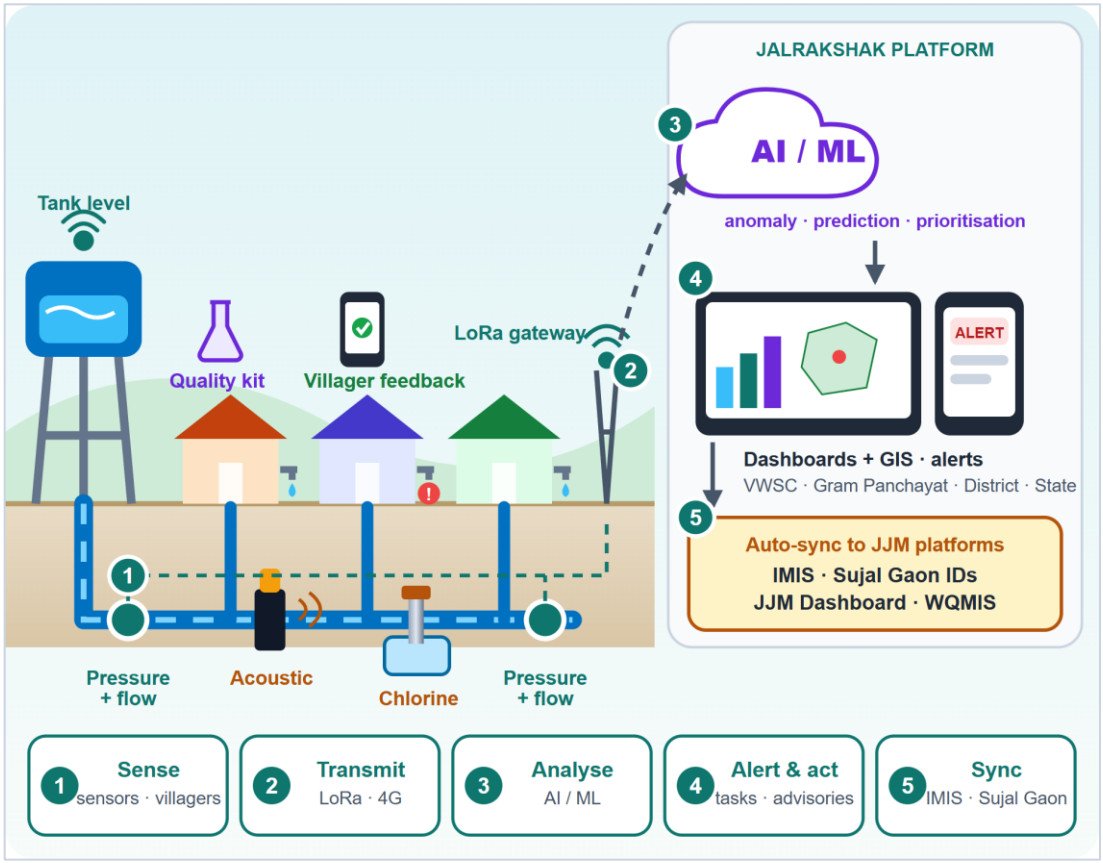
**Found out too late**
Problems surface through periodic reports and field inspections, by which time a tap may be dry for days

**Disruptions nobody sees**
Non-functional taps, intermittent supply, poor water quality, low pressure and failing pipes show up only when someone complains

**Villagers have no easy voice**
No quick way to report a dry tap, share a photo, or confirm that the water really came back

**Data sits in separate places**
Sensors, mobile apps, water-quality results and citizen feedback are rarely seen together in one village view

**A tap is not a service**
98.1% of surveyed homes had a tap, yet only 76.0% were assessed as functional [1]




## OUR SOLUTION: JALRAKSHAK

1. **Continuous FHTC monitoring**
   sensor nodes on the pipeline + a daily villager tap check give household-level status, analysed with AI/ML
2. **Real-time alerts**
   for supply disruption, water quality deterioration, low pressure, leakages and infrastructure failure
3. **Villagers speak up**
   feedback, complaints, photographs and geo-tagged observations from a phone, in local languages
4. **Early detection and prediction**
   AI/ML spots abnormal service patterns early, predicts failures and ranks maintenance
5. **Ready to sync**
   records keyed to Sujal Gaon IDs; IMIS / JJM adapters built, official API access needed
6. **Dashboards for every level**
   GIS maps, service status indicators and analytics from VWSC to State
7. **What is new**
   Beyond sensor-only IoT pilots: HIGH alerts need sensor + villager evidence; repairs close only when readings recover and villagers confirm

**SENSE** Nodes + villagers → **UNDERSTAND** Rules + AI/ML → **ALERT** Alert Center → **REPORT** IMIS / JJM platforms → **FIX** Task + repair → **VERIFY** Readings recover

**DEMO VIDEO (YouTube)**
youtu.be/LYlQtrBbpes

**For:** Village Water & Sanitation Committees (VWSCs), Gram Panchayats, implementing agencies and State authorities.

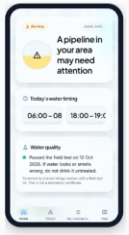
*[1] DDWS, Functionality Assessment of Household Tap Connection, National Report 2024 (weighted sample estimates).*

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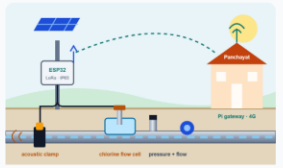
# TECHNICAL APPROACH

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**WHERE DATA STARTS**



Villager phone



Solar sensor node

**SENSE & CAPTURE**

- **Sensors** — pressure · flow · acoustic · chlorine
- **ESP32 + ADS1115** — 4-20 mA loops · MicroPython firmware
- **Villager app** — React PWA · EN / TE / HI
- **Feedback** — photo · geo-tag · daily tap check

→ READINGS + REPORTS

LoRa

**TRANSMIT & INGEST**

- **LoRa SX1276** — 865-867 MHz, node to gateway
- **Raspberry Pi gateway** — store-and-forward buffer
- **4G · MQTT / HTTPS** — device key per node
- **FastAPI ingestion** — validation · rate limit · JWT

→ TRUSTED DATA

MQTT

**AI / ML ANALYSIS**

- **Rules + sensor-health gate** — offline is not "no water"
- **Isolation Forest · Theil-Sen** — anomaly + pressure forecast
- **Chlorine outlook** — repeat-failure window
- **DBSCAN + evidence fusion** — complaint clusters · tap status

→ TAP STATUS + RISK RANK

REST

**ALERT, ACT & SYNC**

- **Alert Center** — acknowledge · escalate
- **Maintenance planner** — task · repair · verified restoration
- **Dashboards + GIS** — VWSC → District → State
- **Auto-sync** — IMIS · Sujal Gaon IDs · JJM · WQMIS

→ ACTION + REPORTING

**TECHNOLOGY STACK**

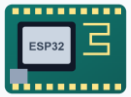
- **APP** — React · Vite · Tailwind; MapLibre · i18next (EN/TE/HI)
- **BACKEND** — Python · FastAPI; SQLAlchemy · JWT
- **DATABASE** — PostgreSQL + PostGIS; SQLite for development
- **AI / ML** — scikit-learn · NumPy; Isolation Forest · DBSCAN · Theil-Sen
- **FIELD / IoT** — ESP32 MicroPython · ADS1115; LoRa SX1276 · Pi · MQTT
- **INTERFACES** — REST + OpenAPI; GeoJSON · CSV
- **DEPLOYMENT** — Docker · Nginx; solar node · IP65 enclosure

**BACKEND AND DATA**

**POSTGRESQL + POSTGIS primary data platform**
Auth (JWT, role-based by village) · photo store (GPS stripped) · audit log · reading retention

**GATEWAY BUFFER fallback**
keeps readings while 4G is down; replays in order

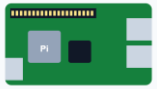
## Proposed hardware and how it connects (firmware tested in simulation; bench build next; circuit on the next slide)




**ESP32 node**
Reads sensors through ADC, I2C and GPIO; sleeps between samples




**LoRa SX1276**
SPI radio, 865-867 MHz, node to village gateway




**Pi gateway**
Buffers readings, sends over 4G MQTT to the ingestion API



**Acoustic sensor**
Piezo clamp at critical points; leak-noise indicator, confirmed by flow and pressure



**Chlorine sensor**
4-20 mA probe in a flow cell at the tail end; measured mg/L, low-chlorine alert




**Pressure x2**
4-20 mA transmitters, head and tail; low-pressure alert



**Flow meter x2**
Pulse output, head and tail; a gap above its normal baseline flags a possible leak



**Ultrasonic level**
JSN-SR04T on the tank; supply and tank health

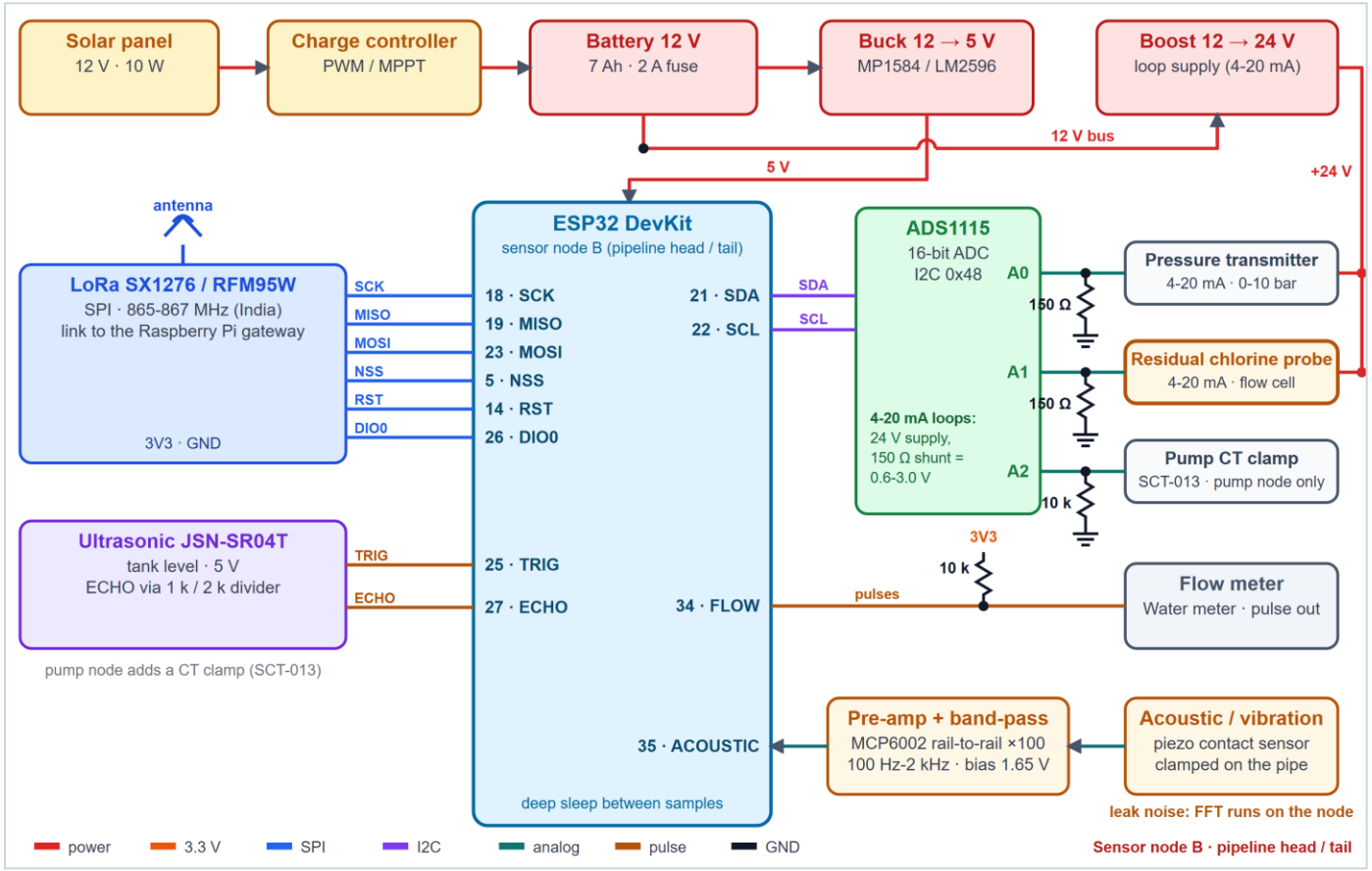



**Solar + battery**
12 V 10 W panel, 7 Ah battery, IP65 box; buck and boost rails

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# FEASIBILITY AND VIABILITY

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*Circuit of sensor node B (pipeline head / tail): ESP32 with LoRa, ADS1115, 4-20 mA loops, flow, acoustic and tank-level inputs.*

## Scalability

- One village, then a cluster per gateway, block, district, state
- Stateless APIs; PostgreSQL + PostGIS
- Same node design in every village

## Interoperability

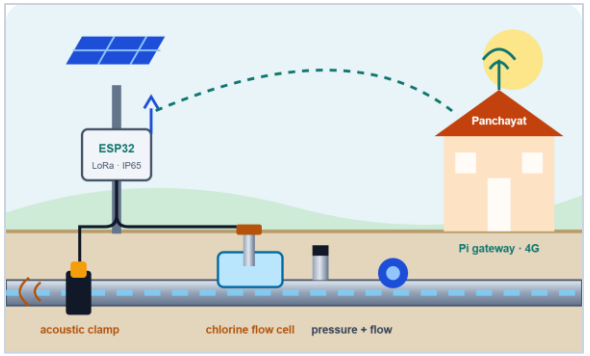
- Keyed to Sujal Gaon and Sujalam Bharat IDs
- Sync adapters built and mock-tested; live IMIS / JJM / WQMIS sync needs official API access
- Open formats: REST/OpenAPI, GeoJSON, CSV

## Data security

- Role-based access by village; hashed passwords
- Photo location data stripped; location shared separately by opt-in
- No personal data sent to other platforms

## Ease of deployment

- Solar node, plug-in LoRa gateway, no grid cabling
- Mobile web app: nothing to install
- Docker deployment, one configuration file




## Node installation concept

- Solar-powered pole node with an IP65 box
- Clamp-on acoustic sensor: no digging
- Chlorine flow cell at the tail end
- LoRa to the Panchayat gateway

## Feasibility analysis

- Off-the-shelf parts: ESP32, ADS1115, LoRa, Raspberry Pi
- Platform on demo data; 278 tests incl. firmware on a simulated board
- AI/ML runs on demo data now; to be validated on pilot data

## Potential challenges and strategies to overcome them

**False alarms:** persistence windows; HIGH needs two kinds of evidence

**Trust in AI:** every score shows its reasons

**Sensor fault or drift:** health gate; chlorine probe checked with a DPD kit

**Leak vs noise:** per-pipe acoustic baseline plus flow gap

**Rural network:** gateway buffers readings, sends when online

**Platform access:** adapters ready; official access and specifications needed

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# IMPACT AND BENEFITS

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## USERS INVOLVED

### Villagers
Report problems with a photo, check their tap daily, and get alerts and advisories in their own language

### VWSC and Gram Panchayat
See alerts, household tap status and a ranked repair list; confirm that water is really restored

### Technicians and operators
Receive tasks with the pipeline, evidence and location; close them when readings recover

### Implementing agencies
Plan preventive maintenance from risk scores and fault history instead of waiting for breakdowns

### District and State authorities
Multi-level analytics and GIS; data prepared for IMIS, Sujal Gaon IDs and the JJM platforms

## SOCIAL BENEFITS

- Dry taps get fixed sooner, and the villager is told when water is restored
- A voice for every household, in local languages
- Boil-water advisory to homes when a field test fails, plus a confirmatory lab test request
- Tail-end homes stop being invisible

## ECONOMIC BENEFITS

- Inspect the riskiest pipes first: fewer repeat repairs and wasted visits
- Leaks near monitored points are flagged earlier
- Less treated water lost to unseen leaks
- Evidence replaces guesswork in planning and budgets

## ENVIRONMENTAL BENEFITS

- Early leak detection saves treated water and pumping energy
- Solar-powered nodes: no grid cabling
- Low-power LoRa links and sleep cycles
- Measured chlorine helps avoid both under- and over-dosing

## EARLY DETECTION AND PREDICTION, THEN ACTION

| DETECT EARLY | PREDICT | ACT BEFORE IT SPREADS |
|---|---|---|
| Acoustic leak noise, pressure trend, chlorine probe and sensor health, before villagers notice | Pressure and chlorine trends, and pipes ranked by recent fault evidence, with reasons | Alert, planned inspection, repair, and restoration verified from readings |

## WHAT A PILOT WILL MEASURE

1. Time from fault to alert, vs complaint-led detection
2. Time from alert to verified restoration
3. Share of days each household reports water
4. False-alarm rate under the two-evidence rule
5. Low-chlorine events caught before a kit test
6. Repairs planned from the ranked risk list

**ROADMAP**

Idea submission → Working software platform → Bench rig with real nodes → Finale demo → Field pilot

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# RESEARCH AND REFERENCES

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## Details / links of the reference and research work


**[1]** DDWS, Functionality Assessment of Household Tap Connection, National Report 2024
jaljeevanmission.gov.in/sites/default/files/2025-12/FHTC_Na...

**[2]** PIB, 13 Aug 2026: 75% of rural habitations covered with Sujal Gaon IDs
pib.gov.in/PressReleasePage.aspx?PRID=2298959

**[3]** PIB, 30 Dec 2025: Jal Seva Aankalan launched
pib.gov.in/PressReleasePage.aspx?PRID=2209798

**[4]** Cabinet, 10 Mar 2026: JJM extended to Dec 2028, service-delivery focus
pmindia.gov.in/en/news_updates/cabinet-approves-extension-of-jal-jeevan-mission...

**[5]** JJM Dashboard (coverage, GIS, IoT pilot view)
ejalshakti.gov.in/JJMReport/JJMIndia.aspx

**[6]** JJM-WQMIS (laboratory and FTK water-quality testing)
ejalshakti.gov.in/WQMIS/

**[7]** PIB, 31 Mar 2021: JJM deploys sensor-based IoT devices (off-grid pilot)
pib.gov.in/PressReleaseIframePage.aspx?PRID=1708701

**[8]** Jal Jeevan Samvad, Oct 2023: IoT water-quality monitoring with complaint tracking
jaljeevanmission.gov.in/sites/default/files/2025-05/JalJeev...

**[9]** Systematic review: ML for anomaly detection in smart water metering networks (2024)
sciencedirect.com/science/article/pii/S1474706524000160

**[10]** SIH 2026 problem statement SIH26255, Ministry of Jal Shakti
sih.gov.in/sih2026PS


## Proposed sensor-node hardware (not yet assembled)


ESP32


LoRa


Pi gateway


Acoustic


Chlorine


Pressure


Flow

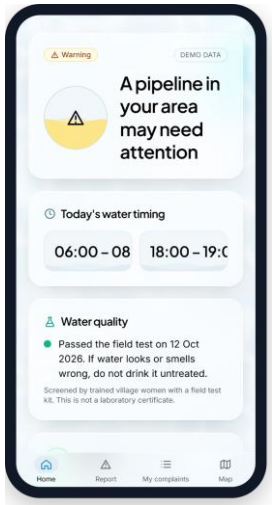

Tank level


Solar

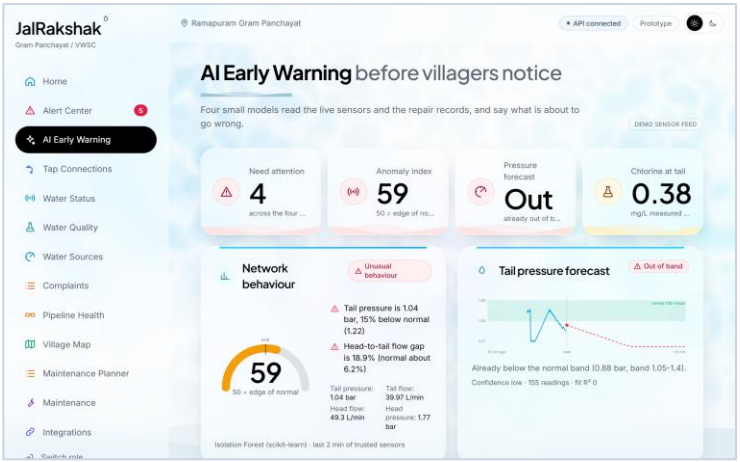
*ESP32-DevKitC + ADS1115, RFM95W LoRa, 4-20 mA pressure and chlorine probes, pulse water meter, piezo clamp, JSN-SR04T. Vendor photos of single parts, not a built prototype; bench build next.*

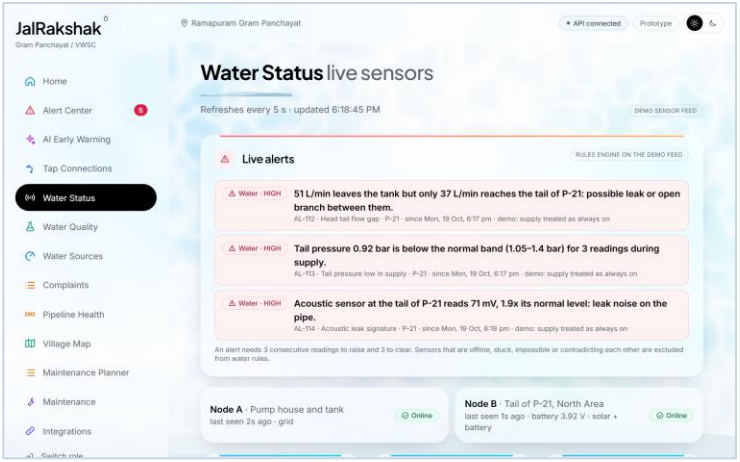
*pH, TDS and turbidity come from field-test-kit entries: screening indicators that trigger confirmatory laboratory testing, not a safety certification.*

## Working platform screens




*Villager app: tap status and water quality (left); report a problem with photo and opt-in location (right)*




**Gram Panchayat:** AI Early Warning with measured chlorine (top right) and Water Status with acoustic leak alerts (bottom right).

**Demo video:** youtu.be/LYlQtrBbpes

**Source code:** github.com/Thrisha236/JalDrishti (platform, node firmware, gateway, circuit and architecture files)